

SFAX THYNA, Tunisia

WMO: 607500

Lat: 34.718N

Lon: 10.691E

Elev: 85

StdP: 14.65

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.7	42.4	25.0	19.2	52.6	28.3	22.3	52.2	24.8	57.0	22.4	56.9	5.3	300	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.6	97.0	72.7	93.6	72.8	91.4	72.9	79.0	87.8	77.8	86.7	76.6	85.7	10.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	140.0	84.2	75.2	132.6	83.4	73.6	125.7	82.8	42.5	88.1	41.2	86.6	40.1	85.8	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.2	18.7	16.5	DB	35.3	107.0	1.6	4.9	34.1	110.5	33.1	113.4	32.2	116.2	31.0	119.8	
			WB	32.3	81.2	2.1	1.8	30.7	82.5	29.5	83.6	28.3	84.6	26.7	85.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.0	54.0	54.9	59.2	64.1	70.4	77.1	81.9	83.0	78.8	72.7	63.2	56.3
	DBStd	11.10	3.49	4.14	4.68	4.29	4.30	4.64	3.54	3.29	4.11	4.62	4.77	4.07
	HDD50	17	8	5	1	0	0	0	0	0	0	0	0	3
	HDD65	1247	341	285	190	65	5	0	0	0	2	88	271	
	CDD50	6605	131	142	285	424	634	813	990	1022	865	704	396	199
	CDD65	2359	0	1	10	39	173	363	525	557	415	241	34	1
	CDH74	23761	2	16	93	206	1050	3436	6327	6902	4031	1547	144	7
	CDH80	9478	0	2	23	38	264	1290	2901	3227	1374	346	13	0
Wind	WSAvg	8.2	7.8	8.3	8.6	9.3	9.2	8.8	8.2	7.8	8.1	7.4	7.5	8.0
Precipitation	PrecAvg	8.9	0.9	0.8	1.0	0.7	0.4	0.2	0.0	0.2	0.9	1.8	0.8	1.1
	PrecMax	25.7	7.2	4.9	2.6	4.1	2.5	1.5	0.6	1.3	4.6	18.2	4.4	6.0
	PrecMin	3.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.4	1.3	0.9	0.7	0.9	0.5	0.3	0.1	0.3	1.1	3.1	0.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	75.9	82.5	84.5	91.8	102.2	103.4	101.4	95.6	91.7	81.0	73.9
		MCWB	55.2	57.5	58.8	62.9	66.9	69.9	72.4	73.4	73.9	69.4	63.4	57.0
	2%	DB	68.0	70.5	75.6	79.2	86.0	93.4	96.5	96.9	91.5	86.2	77.4	70.0
		MCWB	53.7	55.2	58.4	61.5	66.3	69.9	72.1	73.6	74.8	70.1	63.9	56.5
	5%	DB	64.8	67.0	71.8	75.5	82.4	89.6	93.1	93.6	88.9	83.8	75.2	67.8
		MCWB	53.3	54.3	57.5	60.5	65.7	69.7	72.4	74.3	74.0	70.3	63.2	56.1
	10%	DB	62.7	64.3	68.4	73.1	79.2	86.3	90.7	91.3	86.3	80.9	73.1	65.5
		MCWB	52.9	53.4	56.8	59.8	64.5	69.5	72.2	74.3	73.2	69.6	61.8	55.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.8	60.2	63.7	66.9	71.8	77.6	79.6	81.8	80.0	77.0	69.9	62.5
		MCDB	64.3	68.5	73.6	77.7	83.0	89.1	89.1	90.8	87.3	82.5	75.2	67.0
	2%	WB	57.1	58.0	61.4	64.4	69.7	74.7	77.8	79.6	78.3	74.8	67.6	60.1
		MCDB	63.0	65.8	69.8	73.7	80.0	85.4	87.5	88.8	85.4	80.8	73.7	65.5
	5%	WB	55.7	56.4	59.9	62.9	68.2	72.9	76.4	78.2	77.1	73.3	65.5	58.4
		MCDB	61.9	64.0	67.7	71.3	77.6	83.8	86.8	87.7	84.4	79.5	72.2	64.4
	10%	WB	54.2	55.0	58.5	61.6	66.7	71.4	75.2	76.9	75.9	71.7	63.5	56.6
		MCDB	60.9	62.4	65.9	69.7	76.3	82.4	85.8	86.6	83.6	78.6	70.3	63.6
Mean Daily Temperature Range	5% DB	MDBR	18.4	18.4	18.2	16.8	16.6	17.3	17.2	16.6	14.8	15.9	17.9	18.1
		MCDBR	22.0	23.2	23.5	21.4	21.0	22.1	21.3	20.5	16.9	18.4	19.6	20.8
		MCWBR	12.4	12.1	10.9	9.5	8.8	8.9	9.2	8.8	8.0	8.7	10.5	11.7
	5% WB	MCDBR	18.1	19.8	19.0	17.4	17.4	19.1	17.8	17.3	15.0	15.7	16.4	17.0
		MCWBR	11.8	11.6	10.0	8.8	8.5	9.2	8.5	8.4	7.8	8.2	10.4	10.9
Clear-Sky Solar Irradiance	taub	0.356	0.383	0.458	0.509	0.522	0.485	0.511	0.502	0.505	0.461	0.407	0.368	
	taud	2.356	2.258	2.032	1.929	1.925	2.057	2.007	2.037	2.029	2.103	2.205	2.307	
	Ebn at Noon	266	270	257	250	248	257	249	249	240	239	242	252	
	Edh at Noon	31	38	51	60	61	53	56	53	51	43	36	31	
All-Sky Solar Radiation	RadAvg	935	1215	1588	1933	2242	2419	2456	2171	1711	1315	999	837	
	RadStd	49	62	83	99	102	93	56	74	62	68	37	55	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.38	N/A	N/A	N/A	-1.01	-1.35	N/A	N/A	N/A	N/A
Regional (8 neighbors)		+0.43	N/A	N/A	N/A	-0.67	-0.59	N/A	N/A	+200	+104

Nomenclature: See separate page